

SHORT 1

(a)



$F = 1000 \text{ N}$ is added to the piston $\Rightarrow$ the pressure of the gas must increase by

$$p_2 - p_1 = \frac{F}{A} = \frac{1000 \text{ N}}{0.01 \text{ m}^2} = 10^5 \text{ N m}^{-2} \Rightarrow p_2 = 2 \times 10^5 \text{ N m}^{-2}$$

The compression is adiabatic and reversible, and therefore isentropic $\Rightarrow pV^\gamma = \text{const}$

$$\Rightarrow p_1 V_1^\gamma = p_2 V_2^\gamma \Rightarrow \frac{V_2}{V_1} = \left(\frac{p_1}{p_2}\right)^{1/\gamma} \Rightarrow V_2 = 0.01 \times \left(\frac{1}{2}\right)^{3/5} = 6.60 \times 10^{-3} \text{ m}^3$$

(b) By the first law, $\Delta U = Q - W = 0 + p_2(V_1 - V_2) = p_2 V_1 \left(1 - \frac{V_2}{V_1}\right) = 2 \times 10^5 \times 0.01 \times \left(1 - \left(\frac{1}{2}\right)^{3/5}\right) = 680 \text{ J}$

(c) The mass of argon remains unchanged so, by the ideal gas equation of state:

$$p_1 V_1 = mRT_1 = mRT_3 = p_3 V_3 \text{ where } p_3 = p_2 \Rightarrow V_3 = V_1 \frac{p_1}{p_2} = 0.01 \times \frac{1}{2} = 5.00 \times 10^{-3} \text{ m}^3$$

(d) Between states 1 and 3, $\Delta U = C_V \Delta T = 0$ because $T_3 = T_1$. Therefore

$$Q = W = p_2(V_1 - V_3) = p_2 V_1 \left(1 - \frac{V_3}{V_1}\right) = 2 \times 10^5 \times 0.01 \times \left(1 - \frac{1}{2}\right) = 1000 \text{ J}$$

(e) When the gas cools from T_2 to T_3 it contracts so the weight does additional work on the gas, which is dissipated as heat to the environment.

SHORT 2

(a)



By the ideal gas equation of state, $\rho = \frac{p}{RT_1} = \frac{2.87 \times 10^5}{287 \times 10^3} = 1.00 \text{ kg m}^{-3}$

The mass flowrate $\dot{m} = \rho_1 A V_1 \Rightarrow V_1 = \frac{\dot{m}}{\rho_1 A} = \frac{2.00}{2.00 \times 0.01} = 100 \text{ m s}^{-1}$

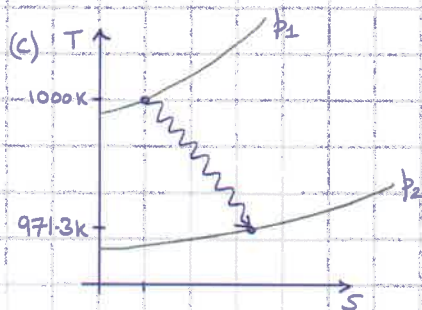
(b) The flow is adiabatic and no shaft work is done on the gas so, by the S.F.E.E.

$$h_1 + \frac{1}{2} V_1^2 = h_2 + \frac{1}{2} V_2^2 \Rightarrow V_2^2 = V_1^2 + 2C_p(T_1 - T_2) = 100^2 + 2 \times 1005 \times 28.66 = 67607$$

$$\Rightarrow V_2 = 260 \text{ m s}^{-1}$$

The mass flowrate is unchanged so $\dot{m} = \rho_1 A V_1 = \rho_2 A V_2 \Rightarrow \rho_2 = \rho_1 \frac{V_1}{V_2} = 1.00 \times \frac{100}{260} = 0.385 \text{ kg m}^{-3}$

By the ideal gas equation of state, $p_2 = \rho_2 R T_2 = 0.385 \times 287 \times (1000 - 28.66) = 1.072 \text{ bar}$



(d) From the perfect gas relationships in the databook, the specific entropy change is:

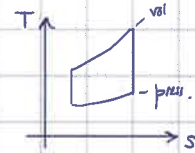
$$s_2 - s_1 = C_p \ln\left(\frac{T_2}{T_1}\right) - R \ln\left(\frac{p_2}{p_1}\right) = 1005 \ln\left(\frac{1000 - 28.66}{1000}\right) - 287 \ln\left(\frac{1.072}{2.87}\right) = 253.4 \text{ J kg}^{-1} \text{ K}^{-1}$$

The mass flowrate is 2.00 kg s^{-1} so the rate of entropy increase is $506.8 \text{ J kg}^{-1} \text{ K}^{-1} \text{ s}^{-1}$

LONG (3)

(a) Process (i) is isentropic and the gas is perfect. Therefore $pV^\gamma = \text{const.}$ and $\frac{T}{p^{\frac{\gamma-1}{\gamma}}}$ is const.

$$\Rightarrow \frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{\frac{\gamma-1}{\gamma}} = r^{2/7} \quad \text{and} \quad \frac{V_2}{V_1} = \left(\frac{p_1}{p_2}\right)^{1/\gamma} = r^{-5/7}$$



(b) Process (ii) is at constant volume. Therefore $Tds = du = c_v dT$

$$\Rightarrow \frac{1}{T} \frac{dT}{ds} = \frac{1}{c_v} \Rightarrow \frac{d(\ln T)}{ds} = \frac{1}{c_v} \quad (\text{check by chain rule of differentiation})$$

If heat is added at constant volume then $W=0$ and, by 1st law, $Q = \Delta U = c_v \Delta T$

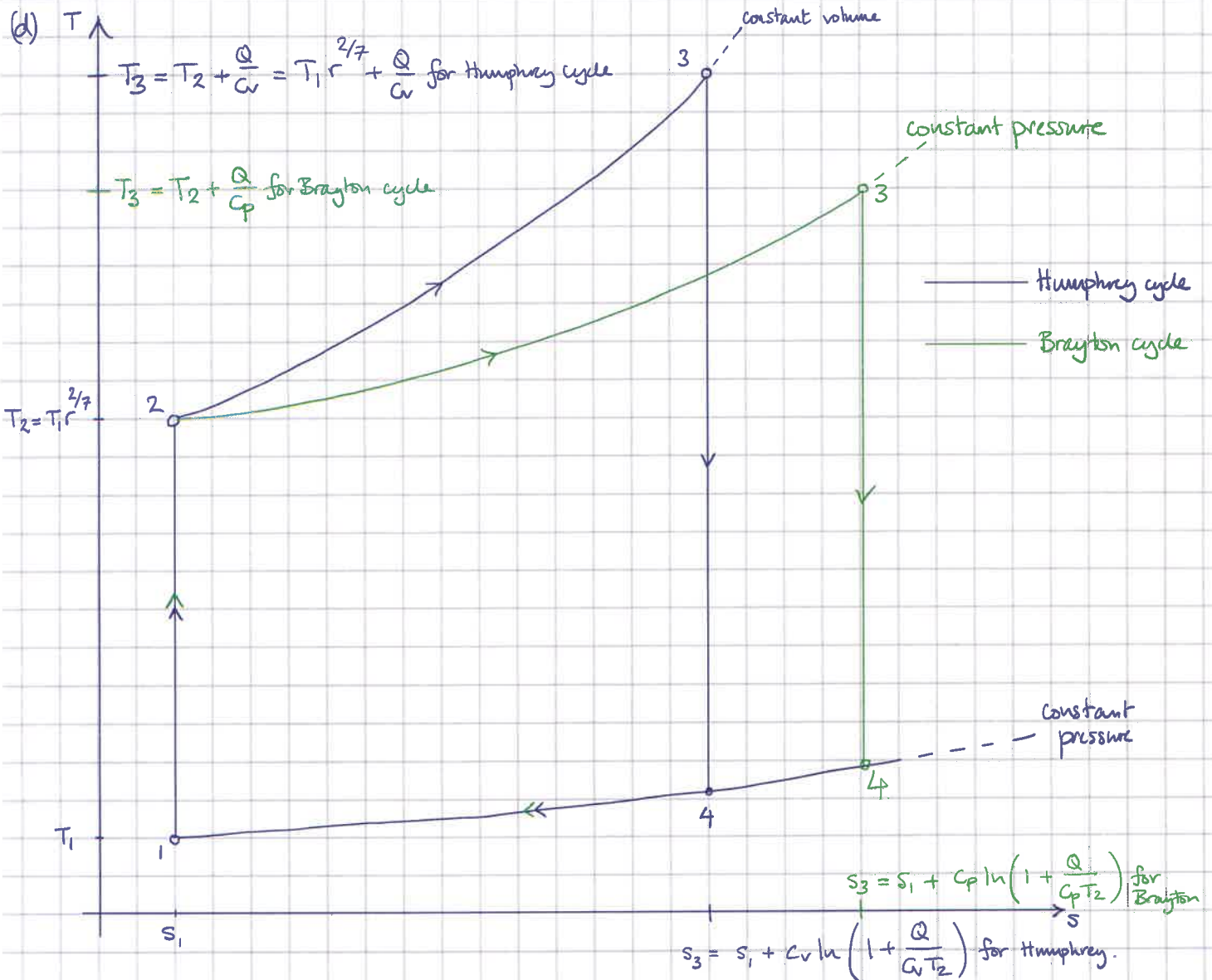
$$\Rightarrow T_3 - T_2 = Q/c_v \quad ; \text{ for next part, } \frac{T_3 - T_2}{T_2} = \frac{T_3}{T_2} - 1 = \frac{Q}{c_v T_2} \Rightarrow \frac{T_3}{T_2} = 1 + \frac{Q}{c_v T_2}$$

Using the perfect gas relationship in the databook or the expression above:

$$s_3 - s_2 = c_v \ln\left(\frac{T_3}{T_2}\right) = c_v \ln\left(1 + \frac{Q}{c_v T_2}\right)$$

(c) Process (iv) is at constant pressure. Therefore $Tds = dh = c_p dT$

$$\Rightarrow \frac{1}{T} \frac{dT}{ds} = \frac{1}{c_p} \Rightarrow \frac{d(\ln T)}{ds} = \frac{1}{c_p}$$



(e) In process (ii) heat is added at constant pressure $\Rightarrow T_3 - T_2 = Q/C_p$

Using the perfect gas relationships, $S_3 - S_2 = C_p \ln\left(\frac{T_3}{T_2}\right) = C_p \ln\left(1 + \frac{Q}{C_p T_2}\right)$

Consider $\frac{d(S_3 - S_2)}{dQ} = \frac{C_p}{1 + \frac{Q}{C_p T_2}} \times \frac{1}{C_p T_2} = \frac{C_p}{C_p T_2 + Q}$ for Brayton and $\frac{C_v}{C_v T_2 + Q}$ for Humphrey.

For all Q , this is always larger for Brayton than for Humphrey, so integrating w.r.t. Q shows that $S_3 - S_2$ of the Brayton cycle is always greater than that of the Humphrey cycle.

(f) The Humphrey cycle is more efficient than the Brayton cycle because:

- (i) for the same heat in, Q , the Humphrey cycle has a larger temperature drop during the isentropic expansion, which is when the work is extracted. By the first law, this will lead to larger work being extracted from the same Q .
- or (ii) for the same heat in, Q , the Brayton cycle has a larger entropy increase $S_3 - S_2$. Consequently, during the heat rejection stage, the Brayton cycle has to reject more heat to the environment, resulting in less work being extracted.
- or (iii) The temperature of heat addition is always greater for the Humphrey cycle, while the temperature of heat rejection is always less or equal to that of the Brayton cycle.

Any of these explanations is sufficient.

(g) $p v^n = \text{const}$. If both p and v increase then n must be negative.

spare questions below
this line. Delete if
not required.

(h) $dQ = T ds = du + p dv = C_v dT + p dv$ so we want to express $p dv$ in terms of dT

$$p v^n = \text{const} \Rightarrow \frac{dp}{p} + n \frac{dv}{v} = 0 \text{ (by product rule)} \Rightarrow v dp = -n p dv$$

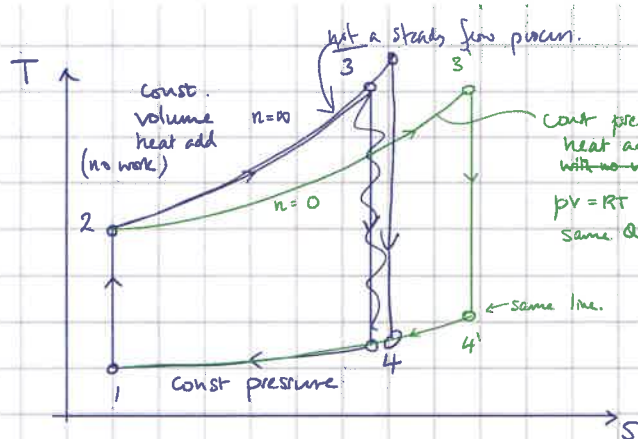
$$p v = R T \Rightarrow p dv + v dp = R dT$$

$$\Rightarrow p dv - n p dv = R dT \Rightarrow (1-n) p dv = R dT \Rightarrow p dv = \frac{R dT}{(1-n)}$$

$$\Rightarrow dQ = C_v dT + \frac{R}{1-n} dT = \frac{C_v(1-n) + C_p - C_v}{1-n} dT = \frac{C_p - n C_v}{1-n} dT$$

$$\Rightarrow T_3 - T_2 = \frac{1-n}{C_p - n C_v} Q \quad \left\{ \begin{array}{l} \text{in limit } n \rightarrow 0, p = \text{const and } T_3 - T_2 \rightarrow Q/C_p \\ \text{in limit } n \rightarrow -\infty, v = \text{const and } T_3 - T_2 \rightarrow Q/C_v \end{array} \right\} \text{ as required.}$$

For negative n , $(T_3 - T_2)$ varies smoothly from that of the Humphrey cycle ($n \rightarrow -\infty$) to that of the Brayton cycle ($n \rightarrow 0$). The efficiency of the real engine will lie somewhere between those of the Humphrey and Brayton cycles.



S.F.E.E. $C_p T = Q$ $C_p(T_3 - T_2) = Q$

$Q = \int T ds$

pulse detonation (not steady flow process).

p increases at const rate v
or v " " " " " " p
both increase at same time $\Rightarrow n$ is negative

if $p v^n = \text{const}$
then, as p increases

$T ds = du + p dv = dh - v dp$

$= du$ if const v
 $= dh$ if const p

if $p v^n = \text{const}$ then $p dv = \frac{R dT}{(1-n)}$ $\Rightarrow T ds = C_v dT + \frac{R}{1-n} dT$

$= \frac{C_v(1-n) + C_p - C_v}{1-n} dT$

$T ds = \frac{C_p - n C_v}{1-n} dT$

$n \rightarrow -\infty$

C_v

$n \rightarrow \infty$ to 0

check: $n=0 \Rightarrow p = \text{const}; T ds = C_p dT$

$n=\infty \Rightarrow v = \text{const}; T ds = C_v dT$

Expression for η as a fn of polytropic eff. n .

what if $n=1$? [cannot?]

$T ds = \infty dT \Rightarrow ds \Rightarrow dT = 0$

const $T \Rightarrow Q_{in} = \text{work out}$
during process
 $\Rightarrow \Delta u = 0, \Delta h = 0$

reheat part

$\frac{dT}{ds} = \frac{1-n}{C_p - n C_v} T$

$\frac{1}{T} \frac{dT}{ds}$

$C_p - n C_v = 0$

$\Rightarrow C_p = n C_v$

$\Rightarrow n = \frac{C_p}{C_v} = \gamma$

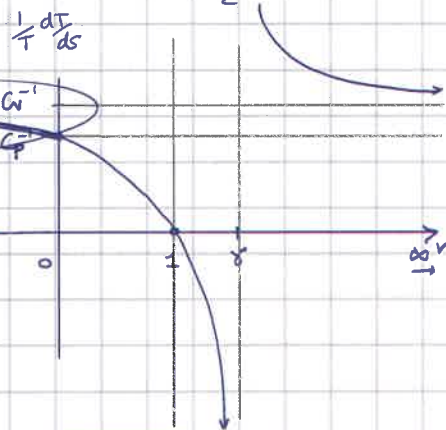
$n \rightarrow 0 \Rightarrow$

$\frac{-(C_p - n C_v) + (1-n) C_v}{(C_p - n C_v)^2} n > \gamma \Rightarrow \frac{1-\gamma}{C_p - \gamma}$ pos

$= -C_p + n C_v + C_v - n C_v$

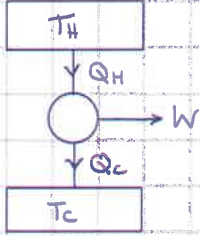
$= -\frac{R}{(\dots)^2}$

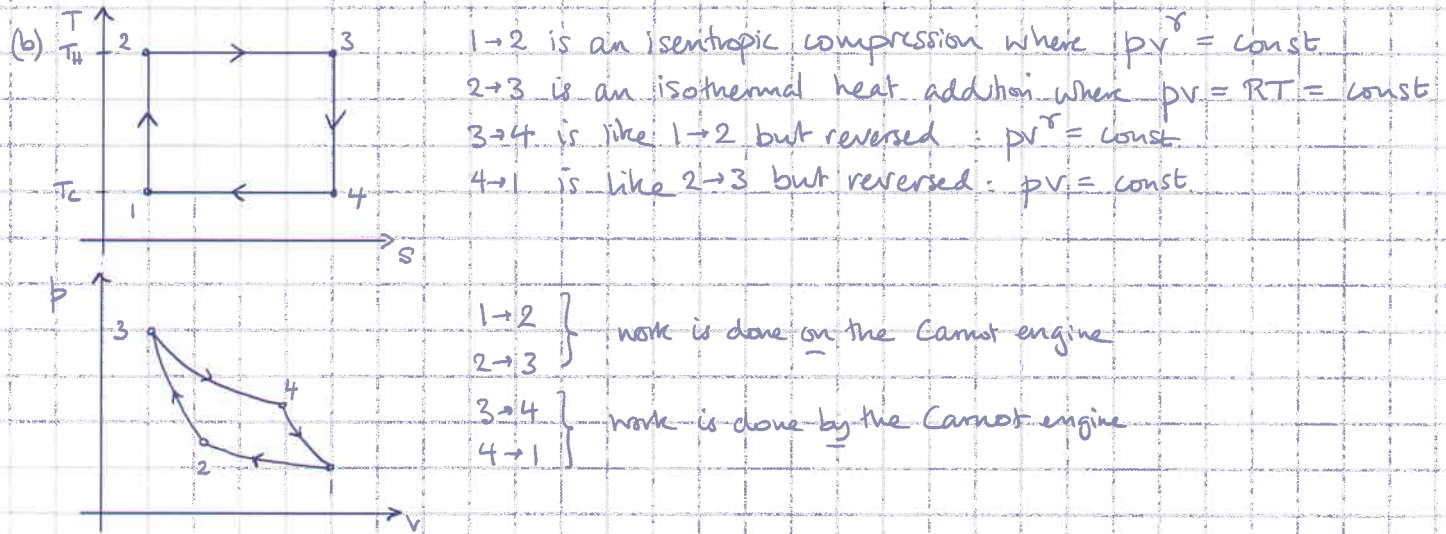
$\frac{1+\epsilon}{C_p - (1+\epsilon) C_v} - \epsilon$



Humphrey cycle: pulse detonation engine; not a steady flow process. $p v^n = \text{const}$ $n = -\infty$ to ∞ to 0.

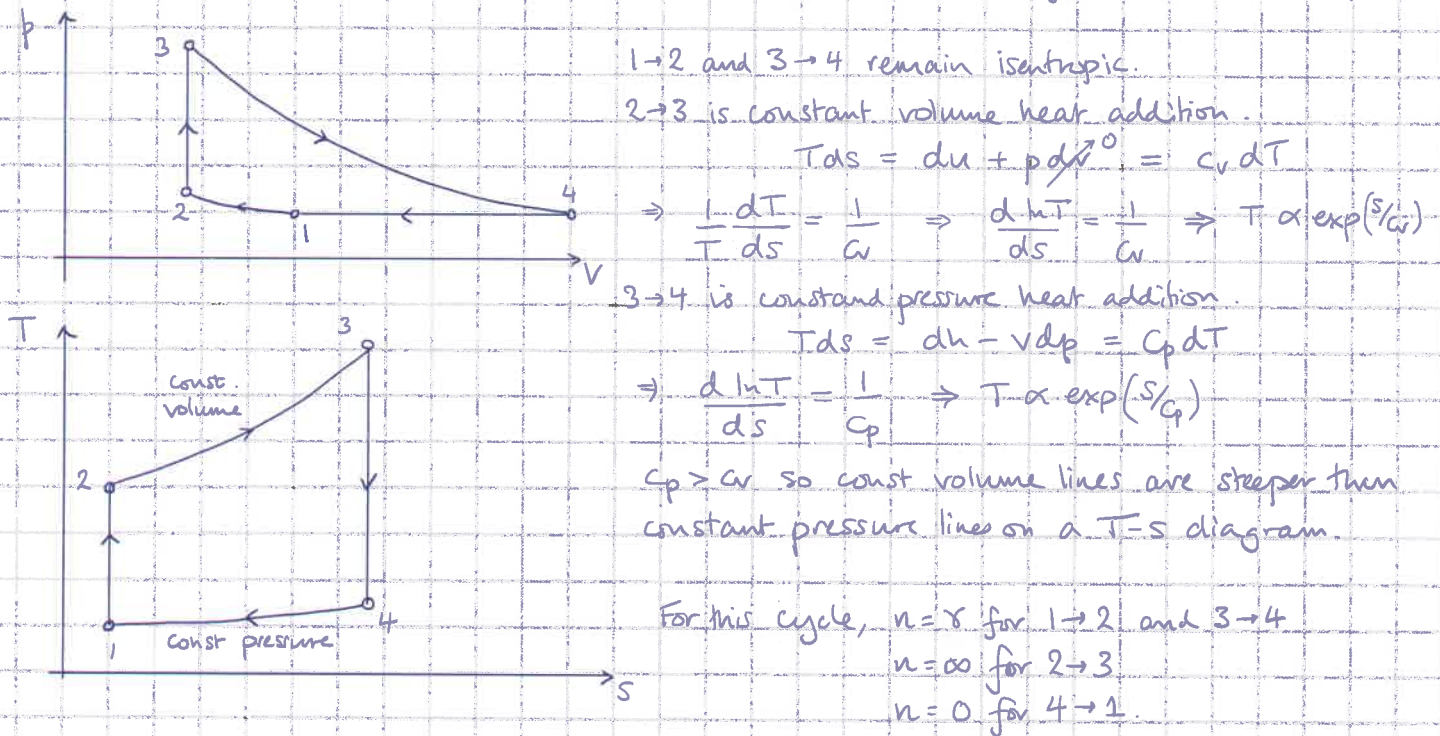
other cycle forms agree to Joule cycle as $n \rightarrow 0$

(a)  1st law: $Q_H = Q_C + W$
 2nd law: $\frac{Q_H}{T_H} - \frac{Q_C}{T_C} \leq 0$ } If reversible, $\frac{Q_H}{T_H} = \frac{Q_C}{T_C}$
 $\Rightarrow \eta \equiv \frac{W}{Q_H} = 1 - \frac{Q_C}{Q_H} = 1 - \frac{T_C}{T_H}$
 maximum efficiency = $1 - T_C/T_H$



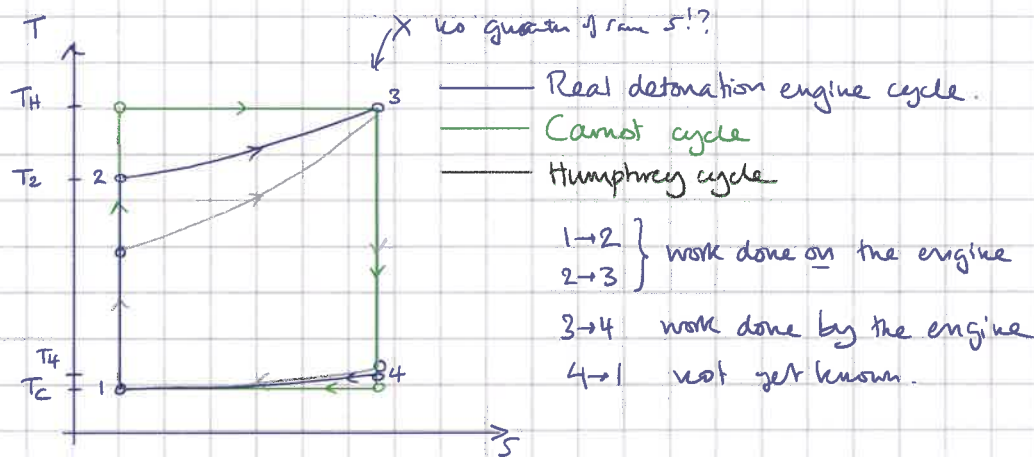
(c) Polytropic process has $pv^n = \text{const}$. For the Carnot cycle, $n = \gamma$ for 1→2 and 3→4
 $n = 1$ for 2→3 and 4→1.

(d) Humphrey cycle: heat added at constant volume and rejected at constant pressure



1→2 work is done on the engine
 2→3 no work done
 3→4 work done by the engine
 4→1 work done on the engine

(e)



$$\frac{d}{d[\cdot]} ab = a \frac{db}{d[\cdot]} + b \frac{da}{d[\cdot]}$$

(f) T_2 is found from $T_3 = T_H$ and $p v^{n_H} = \text{const}$

Ideal gas $\Rightarrow p v = R T$

Also $T ds = du + p dv = dh - v dp$

$T ds = C_v dT + p dv$

$T ds = C_v dT + \frac{R dT}{1 - n_H}$

$T ds = \left(C_v + \frac{R}{1 - n_H} \right) dT$

$= \left(C_v + \frac{C_p - C_v}{1 - n_H} \right) dT$

$= \frac{C_v(1 - n_H) - C_v + C_p}{1 - n_H} dT$

$T ds = \frac{C_p - n_H C_v}{1 - n_H} dT$

$\frac{dT}{T} = \frac{1 - n_H}{C_p - n_H C_v} ds$

nice but not sure it helps.

$n_H = 1 \Rightarrow dT = 0$ ✓

$n_H = \infty \Rightarrow \frac{dT}{T} = \frac{ds}{C_v}$ ✓

$p v = R T$

$p dv + v dp = R dT$

$v^{n_H} \frac{dp}{d[\cdot]} + n_H v^{n_H-1} \frac{dv}{d[\cdot]} p = 0$

$p dv = R dT - v dp$

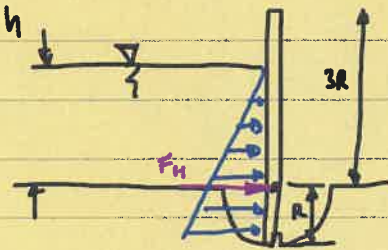
$\Rightarrow \frac{dp}{p} = - n_H \frac{dv}{v}$

$\Rightarrow v dp = - p n_H dv$

$p dv = R dT + p n_H dv$

$p(1 - n_H) dv = R dT \Rightarrow p dv = \frac{R dT}{(1 - n_H)}$

4) a)



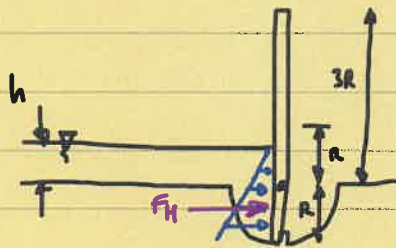
When force acts above
hinge gate will
begin to open.

$$\frac{h+R}{3} > R$$

$$\Rightarrow h > 3R - R > 2R$$

Soap always $\frac{h+R}{3}$ from base so unstable.

b)



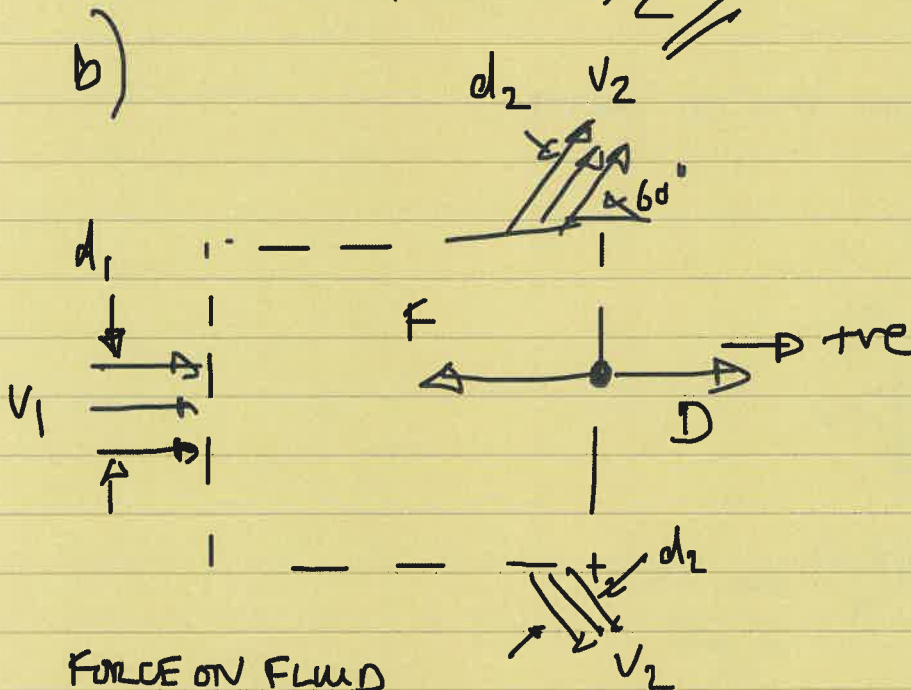
$$F_H = \rho g \left(\frac{R+h}{2} \right)^2 \quad @ \quad \left(\frac{h+R}{3} \right) \text{ from bottom of gate}$$

5) Bernoulli $p_a + \frac{1}{2} \rho v_1^2 = p_a + \frac{1}{2} \rho v_2^2$

a) $\Rightarrow v_1 = v_2$

$d_1 v_1 = 2 \times d_2 v_2$

$\Rightarrow d_2 = d_1 / 2$ BY SYMMETRY



$\sum F_x + \sum (pA)_x = \sum (m v_{out})_x - \sum (m v_{in})_x$

$-F = 2 \times \frac{m}{2} v_2 \cos 60^\circ - m v_1$

$= m v_1 (\cos 60^\circ - 1)$

$+F = m v_1 / 2$

$F = \rho v_1^2 \frac{d_1}{2}$ FORCE ON FLUID

FORCE ON OBSTACLE $D = \rho v_1^2 \frac{d_1}{2}$ TO THE RIGHT

6)

$$a) V_2 = V_1 \frac{\pi D_1^2}{\pi D_2^2} = V_1 \frac{D_1^2}{D_2^2} //$$

$$b) p_1 + \frac{1}{2} \rho V_1^2 = p_2 + \frac{1}{2} \rho V_2^2$$

$$p_1 + \frac{1}{2} \rho V_1^2 = p_2 + \frac{1}{2} \rho V_1^2 \left(\frac{D_1}{D_2} \right)^4$$

$$\Rightarrow p_2 = p_1 + \frac{1}{2} \rho V_1^2 \left\{ 1 - \left(\frac{D_1}{D_2} \right)^4 \right\}$$

LHS RHS

- c)
- | | | | | |
|----|--------------------------|---------------|------------|-----------------------|
| 12 | $p_1 - p_{\text{pitot}}$ | $\rightarrow$ | LHS lower | SAME AS 14 |
| 23 | $p_{\text{pitot}} - p_2$ | $\rightarrow$ | LHS lower | } OPPOSITE |
| 34 | $p_2 - p_{\text{pitot}}$ | $\rightarrow$ | RHS lower | |
| 24 | | $\rightarrow$ | ZERO | (pitot on both sides) |
| 14 | $p_1 - p_{\text{pitot}}$ | $\rightarrow$ | SAME AS 12 | |
| 13 | $p_1 - p_2$ | $\rightarrow$ | LHS lower | |

$$d) i) p_1 - \left(p_1 + \frac{1}{2} \rho V_1^2 \right) = \rho_L g h_{12}$$

$$\Rightarrow h_{12} = - \frac{\rho V_1^2}{2 \rho_L g} //$$

$$ii) p_1 - p_2 = - \frac{1}{2} \rho V_1^2 \left\{ 1 - \left(\frac{D_1}{D_2} \right)^4 \right\} = \rho_L g h_{13}$$

$$h_{13} = - \frac{\frac{1}{2} \rho V_1^2 \left\{ 1 - \left(\frac{D_1}{D_2} \right)^4 \right\}}{\rho_L g} = \frac{\rho V_1^2}{2 \rho_L} \left[\left(\frac{D_1}{D_2} \right)^4 - 1 \right]$$

$$\text{iii) } p_{\text{atm}} - p_2 = \rho_1 g h_{23}$$

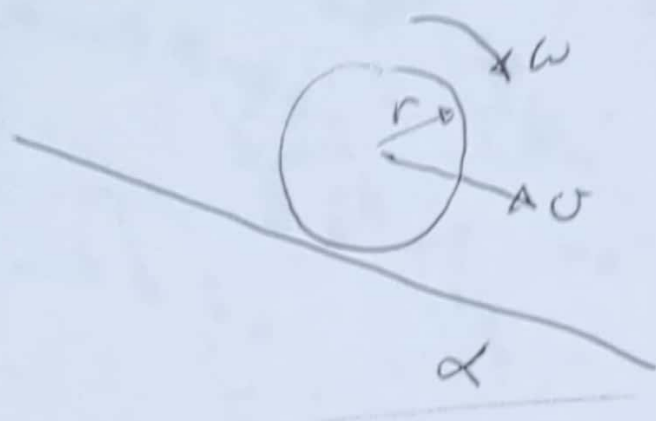
$$\cancel{p_1} + \frac{1}{2} \rho_1 v_1^2 - \left[\cancel{p_1} + \frac{1}{2} \rho_1 v_2^2 \left(1 - \left(\frac{D_1}{D_2} \right)^4 \right) \right] = \rho_1 g h_{23} \Rightarrow \frac{1}{2} \rho_1 v_1^2 \left(\frac{D_1}{D_2} \right)^4 = \rho_1 g h_{23}$$

$$\Rightarrow h_{23} = \frac{\rho_1}{2g\rho_1} v_1^2 \left(\frac{D_1}{D_2} \right)^4 //$$

Mechanics

without slip

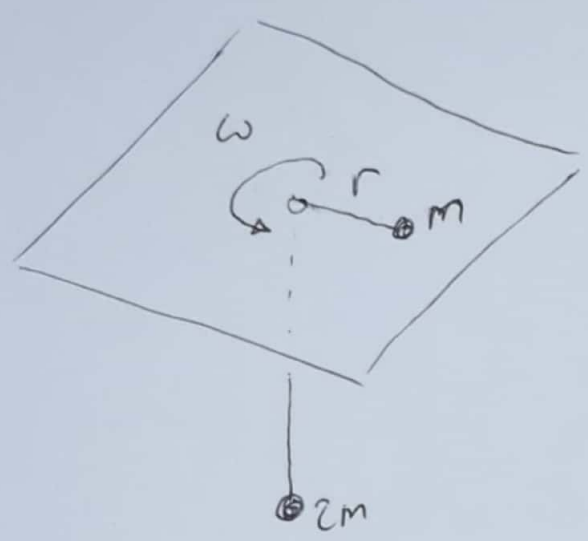
Q6000 A solid sphere is rolling down a ~~flat~~ slope whose angle to the horizontal is α . The sphere has mass m and radius r as shown in Fig 1. The



velocity of the sphere ~~is~~ is v and its angular velocity is ω

- Find an expression for the angular acceleration $\dot{\omega}$ in terms of its acceleration $\dot{v}$
- Show that the normal reaction N between ball and slope is $N = mg \cos \alpha$
- Using d'Alembert's principle or otherwise find an expression for ~~the~~ μ in terms of α for the minimum coefficient of friction μ required for the ball to roll without slip
- For the case $\alpha = 45^\circ$ show that $\mu = \frac{2}{7}$

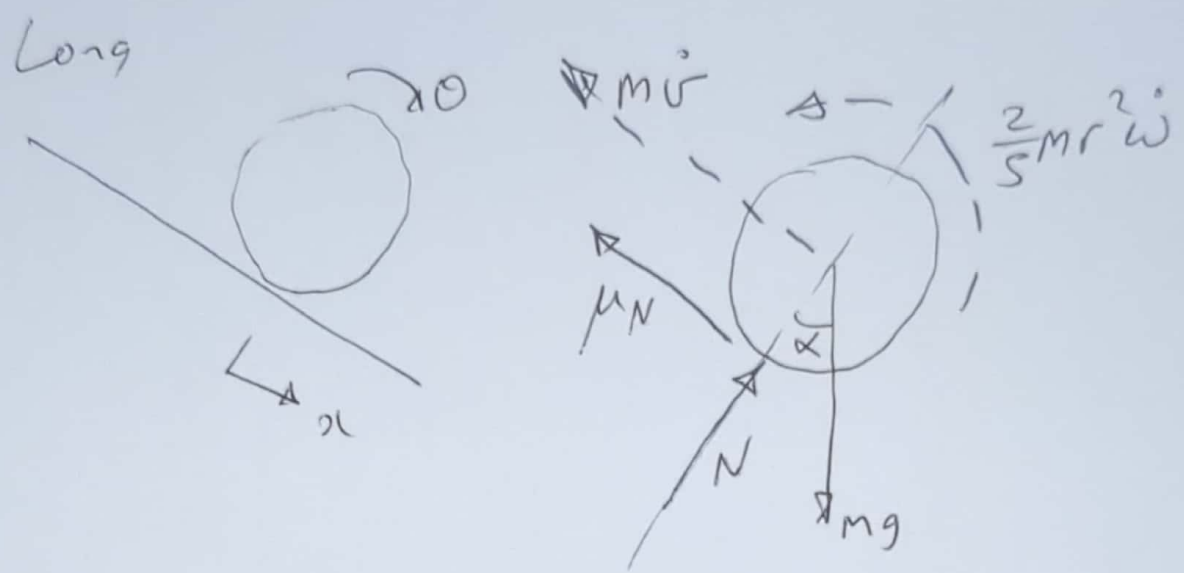
Short



A puck of mass m is sliding on an air table. It is connected ^{by a string} through a hole ^{at O} to a mass of $2m$ as shown in fig 2. Motion of the puck is described by its distance r from O & ~~the~~ its angular velocity ω .

(a) For steady circular motion ^{at radius r} ~~show~~ an expression for ~~Find that~~ the speed of the puck is ~~$v = \sqrt{2gr}$~~

(b) An impulse is applied to the puck when it is moving in a steady circle of radius r . ~~and the radius of the circle increases~~
The impulse ~~is~~ applied in the direction of ~~motion~~ ^{sends the puck to a maximum radius of $2r$} . ~~Find the speed of the puck.~~
~~Find the maximum difference between the minimum speed of the puck~~



(a) rolling $x = r\theta$
 $\dot{x} = r\dot{\theta}$
 $\ddot{x} = r\ddot{\theta} \quad \therefore \dot{v} = r\dot{\omega}$

(b) $\sum F \nearrow \quad \therefore N = mg \cos \alpha$

(c) $\sum F \nearrow \quad \mu N + m\dot{v} = mg \sin \alpha$

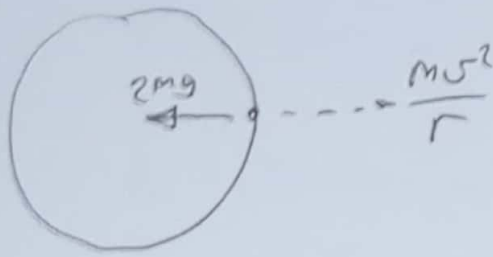
$\sum \text{Moments} \quad \mu N r = \frac{2}{5} m r^2 \dot{\omega} = \frac{2}{5} m r \dot{v}$
 $\therefore \mu N = \frac{2}{5} m \dot{v}$

$\therefore mg \sin \alpha = \mu N \left(1 + \frac{5}{2} \right) = \frac{7}{2} \mu N$

$\therefore \boxed{\mu = \frac{2}{7} \tan \alpha}$

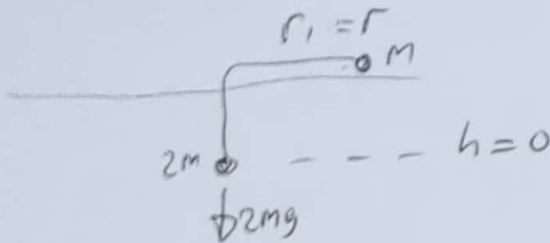
(d) $\alpha = 45^\circ \quad \mu = \frac{2}{7}$

(a)



$$\therefore \omega^2 = 2gr$$

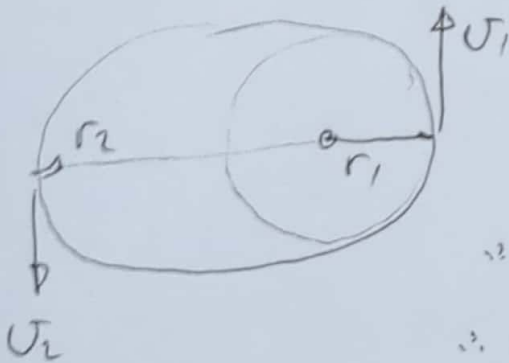
$$\therefore \omega = \sqrt{2gr}$$



(b) Subsequent motion

Angular momentum $r_1 \omega_1 = r_2 \omega_2$

$$\text{Given } \frac{r_2}{r_1} = 2 \therefore \frac{\omega_1}{\omega_2} = 2$$



$$\text{Energy: } \frac{1}{2} m \omega^2 + 2Mg h = \text{const}$$

$$\therefore \frac{1}{2} \omega_1^2 = \frac{1}{2} \omega_2^2 + 2Mg(r_2 - r_1)$$

$$\therefore \omega_1^2 - \omega_2^2 = 4g(r_2 - r_1)$$

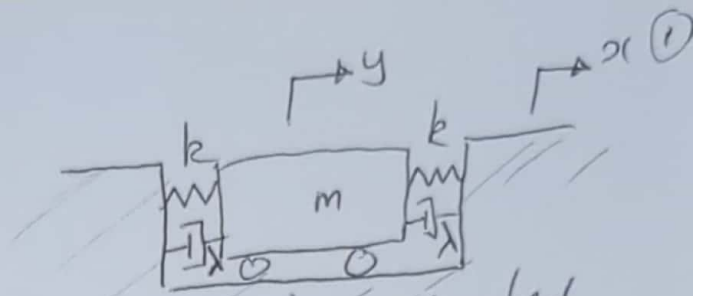
$$\omega_2^2(4 - 1) = 4gr$$

$$\therefore \omega_2^2 = \frac{4gr}{3}$$

$$\therefore \boxed{\omega_2^2 = \frac{4}{3}gr}$$

Short
Ques

Vibration



An electron microscope is situated in the basement of a building. It is vibration-isolated as shown in Fig 1. The microscope is modelled as a rigid mass m whose displacement is y . The basement motion is x and the total isolation stiffness and damping are $2k$ and 2λ respectively.

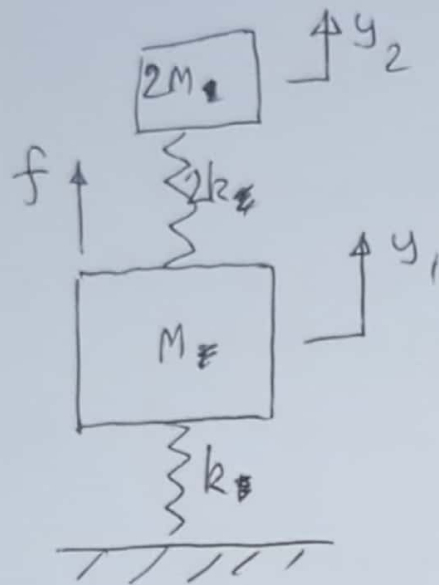
- (a) Find a differential equation for the microscope y given the basement motion x [4]

- (b) Use the data book or otherwise to sketch a graph of the amplitude of motion ~~the~~ y ^{versus frequency} when the basement is moving as $x(t) = X \cos \omega t$.

For your graph use $M = 1000 \text{ kg}$

$k = 10^4 \text{ kN/m}$ and $\lambda = 400 \text{ N/s/m}$
~~10~~

(2)



Long A two-degree-of-freedom system is shown in Fig (2) comprising two masses m and $2m$ described by displacements y_1 and y_2 respectively. The masses are linked by springs k and $2k$ as shown. A force f is applied to mass m .

- (a) Find a matrix equation describing the motion of the system, identifying the mass matrix and the stiffness matrix

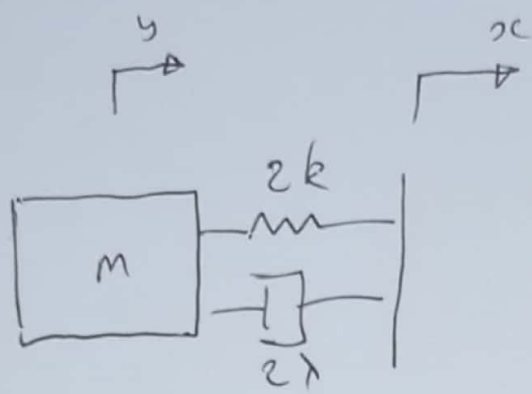
(b) For the case $f(t) = F \cos \omega t$ find
expressions for ~~the~~ Y_1 and Y_2 as a

functions of frequency where $y_1(t) = Y_1 \cos \omega t$
and $y_2(t) = Y_2 \cos \omega t$

(c) sketch graphs of Y_1 & Y_2
vs frequency noting carefully the
resonance frequencies and the
behaviour at $\omega = \sqrt{\frac{k}{m}}$

~~better, show that $Y_1 =$~~

Q185
Short



$m\ddot{y}$
 $2k(y-x) + 2\lambda(\dot{y}-\dot{x})$

 sum of forces = 0
 (d'Alembert)

$$m\ddot{y} + 2\lambda(\dot{y}-\dot{x}) + 2k(y-x) = 0$$

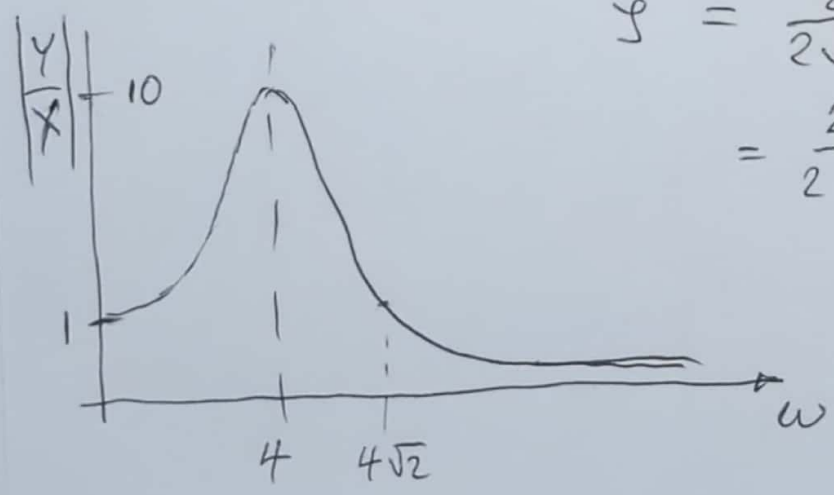
$$\therefore \boxed{m\ddot{y} + 2\lambda\dot{y} + 2ky = 2\lambda\dot{x} + 2kx}$$

This is Case ()

$$\frac{Y}{X} = \frac{1 + 2\gamma \frac{\omega}{\omega_n} i}{1 - \left(\frac{\omega}{\omega_n}\right)^2 + 2\gamma \frac{\omega}{\omega_n} i}$$

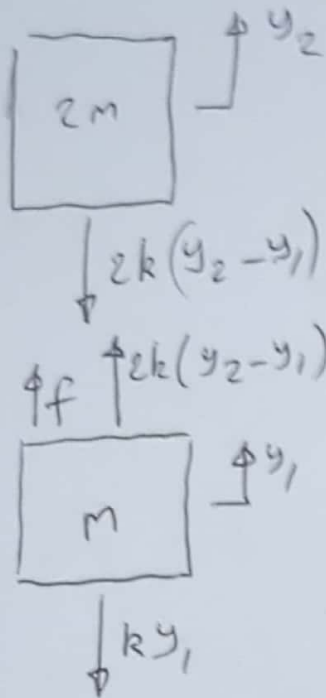
$$\omega_n = \sqrt{\frac{2k}{m}} = \sqrt{\frac{16000}{1000}} = 4 \text{ rad/s}$$

$$\gamma = \frac{2\lambda}{2\sqrt{2km}} = \frac{2 \times 400}{2\sqrt{16000000}} = 0.1$$



ans long

(a)



$$-2k(y_2 - y_1) = 2m\ddot{y}_2$$

$$2k(y_2 - y_1) - ky_1 + f = m\ddot{y}_1$$

$$\therefore \begin{bmatrix} m & 0 \\ 0 & 2m \end{bmatrix} \begin{bmatrix} \ddot{y}_1 \\ \ddot{y}_2 \end{bmatrix} + \begin{bmatrix} 3k & -2k \\ -2k & 2k \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} f \\ 0 \end{bmatrix}$$

$\uparrow$
Mass matrix

$\uparrow$
Stiffness matrix

(b) put $f = F \cos \omega t$, $y_1 = Y_1 \cos \omega t$
 $y_2 = Y_2 \cos \omega t$

$$\begin{bmatrix} 3k - m\omega^2 & -2k \\ -2k & 2k - 2m\omega^2 \end{bmatrix} \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = \begin{bmatrix} F \\ 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = \frac{1}{\Delta} \begin{bmatrix} 2k - 2m\omega^2 & 2k \\ 2k & 3k - m\omega^2 \end{bmatrix} \begin{bmatrix} F \\ 0 \end{bmatrix}$$

(6)

$$\text{where } \Delta = 2(3k - m\omega^2)(k - m\omega^2) - 4k^2$$

$$= 2 \left[3k^2 - 4km\omega^2 + m^2\omega^4 - 2k^2 \right]$$

$$= 2k^2 \left[\left(\frac{\omega}{\omega_n} \right)^4 - 4 \left(\frac{\omega}{\omega_n} \right)^2 + 1 \right]$$

$$\text{where } \omega_n^2 = \frac{k}{m}$$

Two natural frequencies when $[\quad] = 0$

$$\therefore \left(\frac{\omega}{\omega_n} \right)^2 = 2 \pm \frac{1}{2} \sqrt{16 - 4}$$

$$= 2 \pm \sqrt{3}$$

$$Y_1 = \frac{F}{\Delta} 2k \left(1 - \left(\frac{\omega}{\omega_n} \right)^2 \right)$$

$$Y_2 = \frac{F}{\Delta} 2k$$

